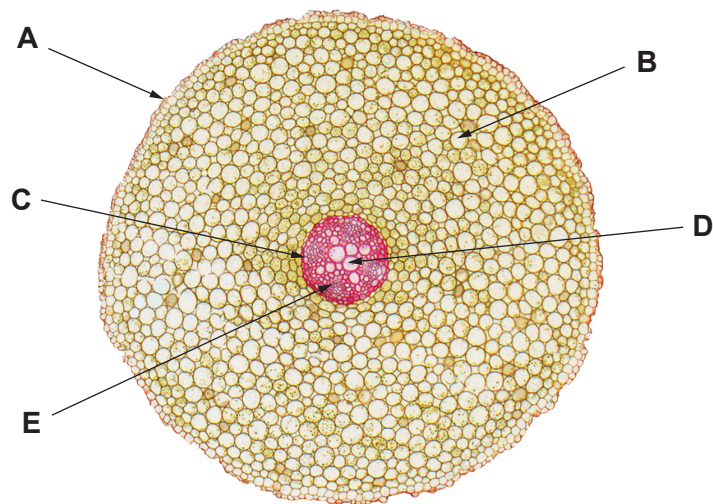


6. Image 6.1 shows a transverse section through a root.

Image 6.1



- (a) Use the **letters (A-E)** from **Image 6.1** to identify the following: [2]
- I. the tissue responsible for transporting organic solutes
 - II. the tissue where root hair cells are located
 - III. the tissue containing the Casparian strip
- (b) Water can be transported across the tissue labelled **B** by the apoplast and symplast pathways. Describe these **two** pathways. [2]

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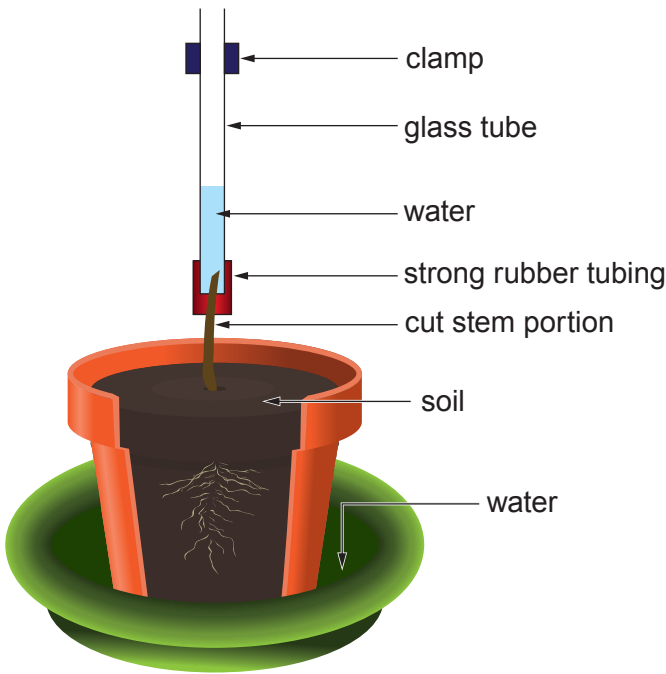
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(c) **Image 6.2** shows an experiment to demonstrate root pressure.

Image 6.2



The stem of a well-watered plant is cut 3 cm above the level of the soil. The cut stem is inserted into a glass tube and secured with rubber tubing to ensure a watertight seal. A small volume of water is poured into the glass tube and its level marked.

After several hours the water level in the tube rises.

- (i) Explain the role of structure **C**, in **Image 6.1**, in the uptake of water into the xylem vessels and in the generation of root pressure. [4]

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(ii) If the same experiment is repeated with cyanide added to the soil, the water level in the tube remains constant. Explain this observation. [3]

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